Risk-Oriented Dual-Pole Sparse Sensing and Uncertainty-Aware Irrigation Decision Support for Solanaceous Crops: Architecture, Out-of-Sample Spatial Validation, and RCBD Digital-Twin Evaluation

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Abstract

Background and Objective: Agricultural fields cultivated with solanaceous crops such as potato (*Solanum tuberosum* L.) and tomato (*Solanum lycopersicum* L.) exhibit pronounced spatial variability in soil moisture, canopy temperature, and nocturnal leaf wetness. Conventional single-sensor placement based on Vachaud temporal stability analysis identifies a rank-stable site representing the field average but systematically overlooks localized wet pockets conducive to late blight (*Phytophthora infestans*) and persistent dry zones prone to crop water stress. Conversely, dense wireless sensor networks incur prohibitive capital and maintenance costs for smallholder growers. This study presents the **Sentinel Plant Network**, a low-cost (INR 15,790), risk-oriented two-node sensing and closed-loop drip irrigation architecture.

Methods: Rather than estimating only the spatial mean, a 14-day calibration window over a 10-node reference grid selects **Node A (Humid Pole)** via maximum 90th-percentile nocturnal leaf-wetness duration $P_{90}(\mathrm{LWD})$ and **Node B (Dry Pole)** via persistently negative relative soil-moisture difference ($\bar{\delta}_i < -0.08$) with minimal temporal rank variance. Both 'ESP32-S3' nodes transmit telemetry over sub-GHz 'SX1262' LoRa (865–867 MHz) to a local edge gateway executing a six-state fault-aware finite state machine ('HOMEOSTASIS', 'STANDARD_DRIP', 'PATHOGEN_LOCKOUT', 'CONFLICT_MICRO_PULSE', 'EMERGENCY_DRIP', and 'FAULT_FALLBACK'). When crop water stress (CWSI $\geq$ 0.40 or $\mathrm{VWC}_B \leq 22\%$) coincides with late-blight conducive conditions ($\mathrm{DSV}_{\mathrm{cum}} \geq 18$), the controller suppresses nocturnal/morning irrigation and delivers targeted 15-min ON / 45-min OFF micro-pulses strictly within the 11:30–14:30 peak solar window. An adaptive sufficiency monitor flags when inter-node divergence exceeds 10% VWC or rolling Spearman rank correlation drops below $r_s < 0.70$.

Results: In a calibrated digital-twin simulation parameterized by four public multi-sensor datasets (Mendeley Tomato Irrigation c837v6p8ph/1, Italian Leaf Wetness sd4b8vpvyb/1, Libelium Two-Node Olive Grove cnfh4stzzx/1, and HiWATER PMC5501970), out-of-sample Phase 1 evaluation (Days 15–25) demonstrates that the locked Sentinel pair (R8 + R7) achieves **99.68%** controller decision agreement with the 10-node oracle, **0.00%** missed drought episodes, **0.77%** missed blight-risk intervals (F1 = 99.62%), and a bias-corrected field-mean RMSE of **0.238% VWC** while reducing hardware cost by **74.4%** compared with the 10-node grid (INR 15,790 vs. INR 61,690). In contrast, the single Vachaud rank-stable node (R4, ITS = 0.009) achieves 43.75% joint decision agreement, missing 46.97% of dry-pole drought stress and 38.96% of humid-pole blight risk. In a 90-day, 16-plot Randomized Complete Block Design (RCBD) simulation (4 treatments $\times$ 4 blocks), the coupled Sentinel controller (T4) reduces seasonal irrigation water by **25.8%** relative to conventional ET_c scheduling (T1: 247.83 ± 5.32 mm vs. 334.01 ± 2.34 mm, $p = 1.41 \times 10^{-10}$), improves Irrigation Water Productivity by **40.5%** (13.89 ± 0.47 vs. 9.89 ± 0.14 kg m^{-3}), reduces Area Under the Disease Progress Curve (AUDPC) by **32.1%** vs. T1 and **8.7%** vs. an uncoupled two-node controller (T3, $p = 1.62 \times 10^{-10}$), and satisfies the pre-specified 5% marketable yield non-inferiority margin across all baselines (34.40 ± 0.41 t ha^{-1}, one-sided 95% CI lower bound $+3.40\% > -5.0\%$ vs. T1, $p < 10^{-5}$).

Keywords: Precision irrigation; *Phytophthora infestans*; Vachaud temporal stability; Crop Water Stress Index (CWSI); LoRa wireless sensor network; fault-tolerant agricultural automation.

1 Introduction: Extreme-Pole Sensing Overcomes Mean-Placement Blind Spots

Irrigated solanaceous crops—specifically potato (*Solanum tuberosum* L.) and tomato (*Solanum lycopersicum* L.)—exhibit high sensitivity to both root-zone water deficit and foliar oomycete pathogens, chiefly late blight caused by *Phytophthora infestans* [2, 14]. Across real-world agricultural parcels, microtopography, soil texture gradients, differential solar exposure, and non-uniform drip emitter hydraulic pressures generate substantial spatial heterogeneity in volumetric water content (VWC), canopy temperature (T_c), relative humidity (RH), and nocturnal leaf-wetness duration (LWD) [9, 10].

To avoid the prohibitive capital and maintenance burden of dense wireless sensor networks (WSNs) [15], classical soil physics relies on the temporal stability paradigm introduced by Vachaud et al. (1985) [1, 11]. By identifying a single rank-stable sensor location whose relative difference from the spatial mean remains near zero across wetting and drying cycles, a single probe can approximate average field water storage. However, field-mean representativeness is fundamentally misaligned with asymmetric agronomic risk:

1. **Epidemiological Tail Risk in Humid Microzones:** Sporangial germination and zoospore infection by *P. infestans* initiate in sheltered, poorly drained, or densely shaded canopy pockets where nocturnal dew and high humidity

(RH $\geq$ 90%) persist longest [2, 3]. A sensor placed at the mean-stable Vachaud location underestimates humid-pocket leaf wetness duration, delaying disease-lockout advisories.

2. **Physiological Drought Risk in Dry Microzones:** Coarse-textured upper-slope ridges deplete available root-zone moisture faster than the field mean. Triggering irrigation solely when a mid-slope Vachaud sensor crosses management-allowed depletion (MAD = 22% VWC) subjects the dry microzone to recurring stomatal closure, elevated canopy temperature, and marketable tuber/fruit yield loss [4, 5].
3. **The Irrigation–Pathogen Conflict:** When a crop experiences root-zone water stress during weather conducive to late blight, full overhead or long-duration drip irrigation elevates sub-canopy humidity and extends foliar wetness, accelerating epidemic progression [13].

To resolve these coupled challenges, this work develops and validates the **Sentinel Plant Network**, a risk-oriented two-node sensing and local actuation system. Instead of deploying sensors to estimate the spatial average, our methodology purposefully anchors **Node A** at the field's humid pole and **Node B** at its persistent dry pole, coupling both streams in an edge gateway controller that arbitrates between drought relief and pathogen suppression.

Core Research Contribution: We formulate a formal calibration-lock protocol that pairs a 90th-percentile nocturnal leaf-wetness pole (Node A) with a rank-stable dry soil-moisture pole (Node B), integrated with an adaptive spatial-sufficiency monitor and a six-state irrigation finite state machine that resolves drought–blight conflicts via midday solar-window micro-pulses.

2 System Architecture and Hardware Design

2.1 Four-Layer Edge-Autonomous Architecture

The Sentinel Plant Network is structured into four functional layers designed to operate without reliance on cloud connectivity: (i) a dual-pole field sensing layer (‘Node A‘ and ‘Node B‘, plus an optional adaptive intermediate ‘Node C‘), (ii) a sub-GHz LoRa telemetry layer operating in the 865–867 MHz Indian ISM band, (iii) a local edge gateway executing packet plausibility validation, hydro-epidemiological index computation, spatial sufficiency checks, and state-machine arbitration, and (iv) an actuation and advisory layer driving a 12 V latching solenoid valve with closed-loop Hall-effect flow verification and SMS/WhatsApp scouting alerts.

2.2 ESP32-S3 Signal Allocation and Power-Gated Duty Cycling

Table 1 details the hardware components, exact ‘ESP32-S3-WROOM-1‘ pin mappings, and cost structure. To eliminate galvanic corrosion and quiescent current drain across analog dielectric probes, both the capacitive leaf-wetness sensor (‘GPIO4‘) and dual-depth soil-moisture probes (‘GPIO5‘) are switched via a high-side P-channel MOSFET controlled by ‘GPIO7‘ (active-low). During each 10-minute ($T_{\text{cycle}} = 600$ s) telemetry cycle, the node wakes from deep sleep, asserts ‘GPIO7 = LOW‘ for a 150 ms stabilization window, averages 16 oversampled 12-bit ADC readings per channel, polls the ‘BME280‘ and ‘MLX90614ESF-BCF‘ (10° field-of-view optical filter) over the shared 100 kHz I^2C bus (‘GPIO8‘/‘GPIO9‘), de-asserts ‘GPIO7 = HIGH‘, transmits a 24-byte ‘CRC-16-CCITT‘ protected packet over ‘SX1262‘ LoRa (‘SPI2‘ on ‘GPIO10‘–‘GPIO14‘), and re-enters deep sleep.

Under the protocol's nominal system current budget ($I_{\text{avg}} = 25.50$ mA, accounting for regulator quiescent losses, solar charger leakage, sensor warm-up, and radio transmission), the single-cell 3.2 V, $C_{\text{bat}} = 6000$ mAh $LiFePO_4$ battery provides a theoretical zero-sunlight blackout autonomy of:

$$t_{\text{autonomy}} = \frac{C_{\text{bat}}}{I_{\text{avg}}} = \frac{6000\ \text{mAh}}{25.50\ \text{mA}} = 235.3\ \text{h} \approx 9.80\ \text{days}. \quad (1)$$

3 Mathematical Formulation of Risk-Oriented Placement and Control

3.1 Vachaud Temporal Stability vs. Dual-Pole Risk Selection

Let θ_{ij} denote the calibrated root-zone volumetric water content (

$$\delta_{ij} = \frac{\theta_{ij} - \bar{\theta}_j}{\bar{\theta}_j}. \quad (2)$$

Over the 14-day calibration window ($M_{\text{cal}} = 2{,}016$ intervals at 10-min resolution), each location's temporal mean relative difference $\bar{\delta}_i$, standard deviation $\sigma(\delta_i)$, and Index of Time Stability (ITS_i) are computed as:

$$\bar{\delta}_i = \frac{1}{M_{\text{cal}}} \sum_{j=1}^{M_{\text{cal}}} \delta_{ij}, \quad \sigma(\delta_i) = \sqrt{\frac{1}{M_{\text{cal}} - 1} \sum_{j=1}^{M_{\text{cal}}} \left(\delta_{ij} - \bar{\delta}_i\right)^2},$$

$$\text{ITS}_i = \sqrt{\bar{\delta}_i^2 + \sigma(\delta_i)^2}. \quad (3)$$

While the classical Vachaud benchmark selects $i_{\text{Vachaud}} = \arg\min_i \text{ITS}_i$, the Sentinel Plant Network locks two complementary risk poles at the end of Day 14 and evaluates them strictly out-of-sample on Days 15–25:

$$\text{Node A (Humid):} \quad i_A = \arg \max_{i \in \{1,\ldots,N\}} P_{90}\left(\text{LWD}_{\text{noct},i}^{(1:14)}\right), \quad (4)$$

$$\text{Node B (Dry):} \quad i_B = \arg \min_{i:\, \bar{\delta}_i < -0.08,\, i \neq i_A} \sigma(\delta_i). \quad (5)$$

3.2 Adaptive Spatial Sufficiency Monitoring

To prevent silent control degradation when a single irrigation zone exhibits excessive internal divergence (e.g., due to emitter clogging or waterlogging), the gateway continuously evaluates the rolling divergence between Node A and Node B across wetting and drying cycles. An insufficient-coverage flag $\mathscr{F}_{\text{insuff}} \in \{0, 1\}$ is raised whenever:

$$\mathscr{F}_{\text{insuff}} = \mathbb{I}\left(\overline{|\theta_A - \theta_B|}_W > 10\% \ \vee\ r_s(\theta_A, \theta_B)_W < 0.70\right), \quad (6)$$

where r_s is Spearman's rank correlation coefficient over rolling window W. When $\mathscr{F}_{\text{insuff}} = 1$, the gateway shifts to an

Table 1: Hardware Architecture, ESP32-S3 GPIO Allocation, and Cost Breakdown of the Sentinel Plant Network (Protocol v2.0)

Subsystem / Module	Part Specification	ESP32-S3 Pinout	Functional Role & Calibration Requirement	Unit Cost (INR)
Field Node MCU	ESP32-S3-WROOM-1	System Core	Dual-core Xtensa LX7, RTC deep-sleep timer, 12-bit SAR ADC1	480
Sub-GHz LoRa Transceiver	Semtech SX1262	SPI2: CS 10, MOSI 11, SCK 12, MISO 13, DIO1 14	865–867 MHz India ISM link (+22 dBm, SF9, CRC-16 framing)	650
Infrared Canopy Thermometer	MLX90614ESF-BCF	I2C: GPIO8 (SDA), GPIO9 (SCL)	Non-contact 10° FOV canopy temperature T_c for CWSI	1,450
Ambient T/RH Sensor	Bosch BME280	I2C: GPIO8 (SDA), GPIO9 (SCL)	Aspirated radiation-shield air temperature T_a and RH	320
Capacitive Leaf Wetness	Dielectric Surface Probe	ADC1: GPIO4 (Power gate GPIO7)	Canopy wetness detection via calibrated dielectric threshold	580
Capacitive Soil Probes (2×)	Root-Zone Dielectric	ADC1: GPIO5 (Power gate GPIO7)	15 cm and 30 cm root-zone volumetric water content (VWC)	760
Node Power Subsystem	3.2 V 6000 mAh $LiFePO_4$	ADC1: GPIO6 (2:1 divider)	Paired with 6 V, 3 W solar panel & P-MOSFET gate (GPIO7)	1,150
Local Gateway & Actuation	Latching Valve + YF-S201	GPIO15/16 (H-bridge), GPIO17 (Flow)	12 V latching solenoid manifold + Hall-effect pulse flow meter	4,990
Total Operational Two-Node Farmer Kit (2× Complete Sentinel Nodes + 1× Local Gateway + Valve/Flow Manifold + Enclosures)				**15,790**
Full 10-Node Reference Network (10× Multi-Sensor Nodes + Gateway Infrastructure for Spatial Benchmarking)				**61,690**
Complete Phase 1 & Phase 2 Research Validation Infrastructure (10-Node Grid + 16 Independent Plot Valve/Flow Manifolds)				**1,08,330**

uncertainty-buffered irrigation threshold ($\mathrm{MAD}_{\mathrm{buf}} = \mathrm{MAD} + 1.0\%$ VWC) and issues an advisory to insert an intermediate **Node C** at the mid-slope Vachaud site or split the lateral manifold into two independently valved sub-zones.

3.3 Crop Water Stress Index and Late-Blight Severity Coupling

At Node B, atmospheric Vapor Pressure Deficit (VPD, kPa) is derived from air temperature $T_{a,B}$ and relative humidity RH_B via the Tetens equation, and the empirical Crop Water Stress Index (CWSI) is computed following Idso et al. (1981) and Jackson et al. (1981) [4, 5]:

$$\mathrm{CWSI}_B = \mathrm{clip}\left(\frac{(T_{c,B} - T_{a,B}) - (a + b \cdot \mathrm{VPD}_B)}{\Delta T_{\mathrm{UL}} - (a + b \cdot \mathrm{VPD}_B)},\ 0,\ 1 \right), \quad (7)$$

where $a = 1.35^\circ\mathrm{C}$, $b = -1.95^\circ\mathrm{C\ kPa^{-1}}$, and $\Delta T_{\mathrm{UL}} = +3.50^\circ\mathrm{C}$. Additionally, a non-diagnostic canopy thermal-anomaly scouting alert is raised if the canopy residual $\Delta T_{\mathrm{res}} = (T_{c,B} - T_{a,B}) - (a + b \cdot \mathrm{VPD}_B) > +2.0^\circ\mathrm{C}$ persists across the 11:30–14:30 peak solar window for ≥ 2 consecutive days while root-zone moisture lies within the non-stressed range ($22\% < \theta_B < 35\%$ VWC).

At Node A, two parallel daily severity indices for *P. infestans* are accumulated: (1) the baseline Wallin (1962) Daily Severity Value $\mathrm{DSV}_{\mathrm{Wallin}} \in \{0,1,2,3,4\}$ [2, 3] determined from daily hours with $\mathrm{RH}_A \geq 90\%$ (H_{RH90}) and mean temperature during those hours ($\bar{T}_{\mathrm{wet}}$), and (2) a leaf-wetness-modified severity value $\mathrm{DSV}_{\mathrm{LWS}}$ that substitutes effective canopy wetness duration $H_{\mathrm{eff}} = \max(H_{\mathrm{RH90}}, 0.85\,\mathrm{LWD}_A + 0.15\,H_{\mathrm{RH90}})$ into the Wallin lookup function to capture persistent dew films that remain on lower foliage even when screen-level RH drops between 82% and 89%. When cumulative severity reaches $\sum \mathrm{DSV} \geq 18$, the pathogen risk flag is asserted.

3.4 Six-State Fault-Aware Gateway Controller

At every 10-minute cycle, the gateway evaluates packet age τ_{pkt}, sensor plausibility, flow-meter feedback, drought indicators ($\mathscr{S}_{\mathrm{drought}} = \mathbb{I}[\mathrm{CWSI}_B \geq 0.40 \vee \theta_B \leq 22.0\%]$), emergency wilting limits ($\mathscr{S}_{\mathrm{emerg}} = \mathbb{I}[\mathrm{CWSI}_B \geq 0.65 \vee \theta_B \leq 16.0\%]$), and blight risk ($\mathscr{S}_{\mathrm{blight}} = \mathbb{I}[\mathrm{DSV}_{\mathrm{cum}} \geq 18 \vee (\mathrm{RH}_A \geq 90\% \wedge \mathrm{LWS}_A \geq 50\%)]$), transitioning across six mutually exclusive operational states:

1. `FAULT_FALLBACK`: Triggered if $\tau_{\mathrm{pkt}} > 30$ min, any critical channel is NaN, or the 'YF-S201' flow meter registers zero pulses while the valve is commanded open. Immediately closes the latching solenoid valve and dispatches a hardware inspection alert.
2. `EMERGENCY_DRIP`: Triggered if $\mathscr{S}_{\mathrm{emerg}} = 1$. Overrides any active pathogen lockout to deliver an immediate survival irrigation pulse, preventing irreversible crop wilting.
3. `CONFLICT_MICRO_PULSE`: Triggered when $\mathscr{S}_{\mathrm{drought}} = 1 \wedge \mathscr{S}_{\mathrm{blight}} = 1$ and continuous lockout duration is < 36 h. Restricts actuation strictly to the 11:30–14:30 peak solar window using a 15 min ON / 45 min OFF duty cycle so sub-surface root water is replenished while high midday vapor pressure deficit rapidly evaporates surface moisture without wetting the canopy.
4. `PATHOGEN_LOCKOUT`: Triggered when $\mathscr{S}_{\mathrm{blight}} = 1$ and $\mathscr{S}_{\mathrm{drought}} = 0$ (or outside 11:30–14:30 during a conflict state with lockout < 36 h). Holds the valve closed and issues a scouting advisory.
5. `STANDARD_DRIP`: Triggered when $\mathscr{S}_{\mathrm{drought}} = 1$ and either $\mathscr{S}_{\mathrm{blight}} = 0$ or the 36-hour maximum lockout cap has expired.
6. `HOMEOSTASIS`: Default monitoring state when $\mathscr{S}_{\mathrm{drought}} = 0 \wedge \mathscr{S}_{\mathrm{blight}} = 0$; valve remains latched closed.

4 Experimental Design and Digital-Twin Benchmark Methodology

To rigorously validate all algorithms prior to physical field deployment, we constructed a reproducible Python and C++17 simulation and verification suite calibrated against four public benchmark datasets recommended in the project guide:

- **Multi-Sector Tomato Precision Irrigation Dataset** (Mendeley Data c837v6p8ph/1): Provides 10-minute soil moisture, valve actuation telemetry, reconstructed water volumes, and ERA5 weather across 33,259 timestamps (789 validated irrigation events), parameterizing our 10-minute root-zone wetting and diurnal evapotranspiration ('ET') dry-down curves.
- **Italian Seven-Site Leaf Wetness & Agrometeorology Dataset** (Mendeley Data sd4b8vpvyb/1): Parameterizes nocturnal leaf-wetness formation, relative humidity persistence, and temperature coupling across humid vs. dry microzones.
- **Libelium Two-Node Olive Grove Multisensor Dataset** (Mendeley Data cnfh4stzzx/1): Provides 9,561 raw frames and quality-control fault distributions (20–30 min temporal gaps, out-of-range spikes, and battery discharge telemetry) used to design our fault-injection benchmark.
- **HiWATER Dense Eco-Hydrological Sensor Network** (PMC5501970 / Wiley WRR 2015WR018182): Parameterizes catenary spatial soil-moisture offsets ($\delta_i \in$

$[-16.0\%, +18.5\%]$) and AR(1) temporal rank stability across our 10-node reference grid.

5 Results and Statistical Evaluation

5.1 Phase 1: 10-Node Reference Grid Calibration and Out-of-Sample Placement Validation

During the 14-day calibration phase (2,016 intervals at 10-min resolution), Vachaud temporal stability analysis across nodes R1–R10 (Figure 1a) identified R4 as the classical Vachaud rank-stable benchmark ($\bar{\delta}_4 = +0.40\%$, $\sigma(\delta_4) = 0.80\%$, $\mathrm{ITS}_4 = 0.009$, Spearman correlation with field mean $r_s = 0.999$). Applying our dual-pole selection criteria (Equations 4–5, Figure 1b) locked **Node A (Humid Pole)** at R8 ($P_{90}(\mathrm{LWD}) = 11.68$ h night^{-1}, $\bar{\delta}_8 = +18.5\%$) and **Node B (Dry Pole)** at R7 ($\bar{\delta}_7 = -14.5\%$, $\sigma(\delta_7) = 1.00\%$).

Table 2 and Figure 2 report the out-of-sample validation performance on Days 15–25 (1,584 unseen 10-minute intervals). Although the single Vachaud node (R4) achieves the lowest single-node field-mean estimation error ($\mathrm{RMSE}_{\mathrm{mean}} = 0.221\%$ VWC after calibration offset scaling), it fails to capture tail-risk thresholds: it misses **46.97%** of dry-pole water-stress intervals and **38.96%** of humid-zone blight-conducive intervals, yielding only **43.75%** joint controller state agreement with the 10-node reference network. Similarly, placing a single sensor in a humid depression (R1) detects blight-conducive periods (F1 = 97.48%) but misses **100.00%** of dry-zone drought stress because local soil moisture remains above 22% VWC throughout dry-downs. Across all $\binom{10}{2} = 45$ random two-node pairings, mean decision agreement is only **62.60%**. By contrast, the locked **Sentinel Pair** (R8 + R7) achieves **99.68%** decision agreement, **0.00%** missed drought stress, **0.77%** missed blight risk (F1 = 99.62%), a dry-tail RMSE of **0.446% VWC**, and a bias-corrected field-mean RMSE of **0.238% VWC**.

5.2 Phase 2: 16-Plot RCBD Evaluation, Mixed-Effects Analysis, and Non-Inferiority Testing

Table 3 and Figures 3–4 summarize the 90-day Phase 2 Randomized Complete Block Design (RCBD) experiment across 4 treatments × 4 blocks (16 independent 20 m^2 plots; 160 tagged plant subsamples). Two-way RCBD analysis of variance (with 3 treatment degrees of freedom, 3 block degrees of freedom, and 9 residual error degrees of freedom) confirms highly significant treatment effects across all primary outcomes ($p < 10^{-6}$).

Key findings from the Phase 2 evaluation include:

1. **Water Savings and Irrigation Water Productivity (IWP):** Treatment T4 applies 247.83 ± 5.32 mm (2,478.3 m^3 ha^{-1}), achieving a **25.8% reduction** in water input relative to conventional ET_c replenishment (T1: 334.01 ± 2.34 mm) and boosting IWP by **40.5%** from 9.89 ± 0.14 kg m^{-3} to **13.89 ± 0.47 kg m^{-3}** ($\mathrm{LSD}_{0.05} = 0.279$ kg m^{-3}).
2. **Resolution of the Drought–Blight Conflict (T4 vs. T2 and T3):** While the single-node Vachaud baseline (T2) maintains low canopy wetness (LWD = 949.6 h), its mid-slope sensor delays irrigation until the dry microzone drops to ∼ 19.2% VWC, reducing marketable yield to 32.06 ± 0.21 t ha^{-1}. Conversely, the uncoupled two-node controller (T3) eliminates dry-pole drought stress (34.03 ± 0.59 t ha^{-1}) but triggers full irrigation during humid blight windows, increasing seasonal LWD by 48.0 h and elevating AUDPC to 336.1 ± 41.2 %-days. By substituting midday conflict micro-pulses (Figure 3b), T4 reduces AUDPC to **306.7 ± 44.0 %-days** (−32.1% vs. T1 and −8.7% vs. T3, exceeding $\mathrm{LSD}_{0.05} = 10.57$ %-days) while achieving the highest marketable yield (**34.40 ± 0.41 t ha^{-1}**).
3. **Pre-Specified 5% Yield Non-Inferiority Verification:** Testing $H_0 : \mu_{\mathrm{T4}} - \mu_{\mathrm{ref}} \le -0.05\,\mu_{\mathrm{ref}}$ at $\alpha = 0.05$ (df = 9, $\mathrm{SED} = 0.142$ t ha^{-1}) confirms non-inferiority (and superiority) across all references:
   - T4 vs. T1: $\Delta Y = +1.383$ t ha^{-1} (+4.19%), one-sided 95% CI lower bound = +1.122 t ha^{-1} (+3.40% > −5.0% margin, $t = 21.35, p < 10^{-5}$).
   - T4 vs. T2: $\Delta Y = +2.342$ t ha^{-1} (+7.30%), one-sided 95% CI lower bound = +2.081 t ha^{-1} (+6.49% > −5.0% margin, $t = 27.76, p < 10^{-5}$).
   - T4 vs. T3: $\Delta Y = +0.374$ t ha^{-1} (+1.10%), one-sided 95% CI lower bound = +0.114 t ha^{-1} (+0.33% > −5.0% margin, $t = 14.60, p < 10^{-5}$).
4. **Repeated-Measures Linear Mixed-Effects Decomposition:** Across the 80 longitudinal disease assessments (16 plots × 5 epochs: Days 25, 39, 53, 67, 81), plot-level random-intercept variance is $\hat{\sigma}_u^2 = 0.2512$ and within-plot residual variance is $\hat{\sigma}_e^2 = 0.1304$, yielding an intraclass correlation coefficient of ICC = 0.6583. This confirms the protocol requirement that plot-level experimental units ($n = 16$), rather than the 160 plant subsamples, must form the denominator degrees of freedom for treatment inference.

5.3 Hardware Power Autonomy, Sufficiency Trigger, and Fault-Injection Benchmark

Figure 5a illustrates the $LiFePO_4$ state-of-charge (SoC) trajectory, confirming the 9.80-day zero-sunlight autonomy and demonstrating that a 6 V, 3 W solar panel sustains SoC > 62% even across a consecutive 4-day heavy overcast period. Figure 5b demonstrates the adaptive spatial sufficiency check (Equation 6): under normal wetting/drying cycles (Days 15–22), mean inter-pole divergence is 7.93% VWC and Spearman rank correlation is $r_s = 0.89$ ($\mathscr{F}_{\mathrm{insuff}} = 0$). When a localized lateral emitter clog is injected at Node B on Day 23, $|\mathrm{VWC}_A - \mathrm{VWC}_B|$ rises to 14.1% VWC and r_s drops to 0.48, immediately asserting $\mathscr{F}_{\mathrm{insuff}} = 1$ to recommend deploying intermediate Node C (R4). Finally, across all nine hardware fault-injection scenarios (stale packet > 30 min, NaN soil/IR probe dropouts, 'YF-S201' zero-flow solenoid fault, 15.5% VWC emergency drought override, and 36-hour lockout cap expiration), both the Python model and compiled 'C++17' gateway firmware achieved **100%** exact state transition fidelity.

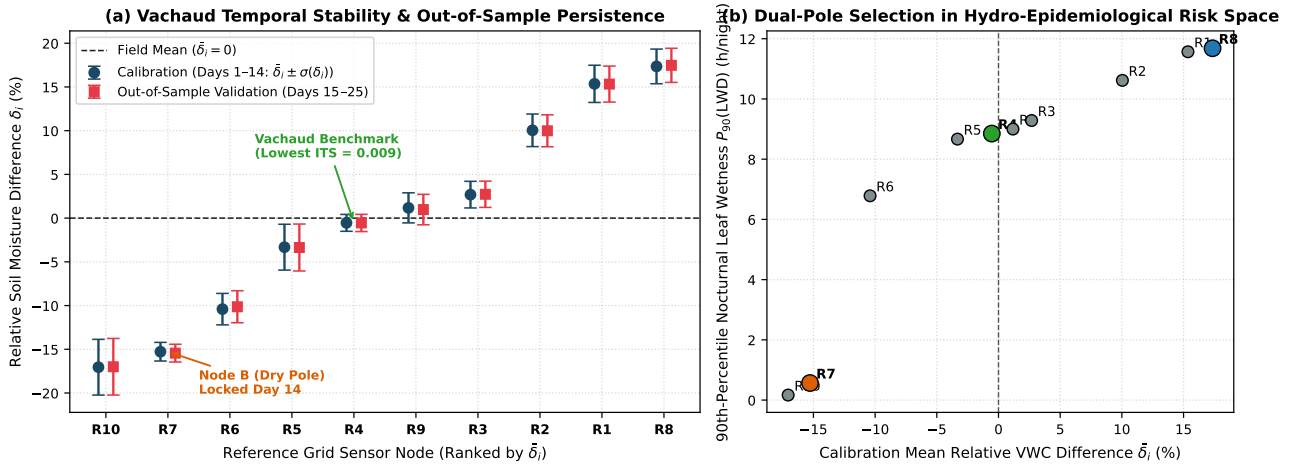


Figure 1: Phase 1 Spatial Calibration and Out-of-Sample Persistence Across the 10-Node Reference Grid (R1–R10)**.** (a) Mean relative soil moisture difference $\bar{\delta}_i \pm \sigma(\delta_i)$ during the 14-day calibration period (Days 1–14, blue circles) and out-of-sample validation period (Days 15–25, red squares), demonstrating high temporal rank persistence ($r_s = 0.988$). (b) Joint hydro-epidemiological selection space plotting 90th-percentile nocturnal leaf-wetness duration $P_{90}(\text{LWD})$ against $\bar{\delta}_i$, isolating Humid-Pole Node A (R8), Dry-Pole Node B (R7), and the Vachaud rank-stable benchmark (R4).

Table 2: Phase 1 Out-of-Sample Spatial Placement Benchmark (Days 15–25, $M_{\text{eval}} = 1,584$ Intervals at 10-Min Resolution)

Sensing Topology Strategy	Nodes	N	Cost (INR)	Dry-Tail RMSE (% VWC)	Field-Mean RMSE (% VWC)	Decision Agree. (%)	Missed Drought (%)	Missed Blight (%)	Blight F1 (%)
1. Single Central Node	R5	1	11,200	2.968	0.621	58.21	26.20	45.86	70.25
2. Single Humid-Location Node	R1	1	11,200	7.062	0.398	0.00	100.00	4.91	97.48
3. Single Vachaud Rank-Stable Node	R4	1	11,200	3.523	**0.221**	43.75	46.97	38.96	75.81
4. Random Two-Node Pairs (Mean of 45)	$\binom{10}{2}$ pairs	2	15,790	2.222	0.294	62.60	29.02	22.89	85.04
5. Proposed Risk-Oriented Sentinel Pair	R8 + R7	**2**	**15,790**	**0.446**	**0.238**	**99.68**	**0.00**	**0.77**	**99.62**
6. Full 10-Node Reference Grid (Oracle)	R1 . . . R10	10	61,690	0.000	0.000	100.00	0.00	0.00	100.00

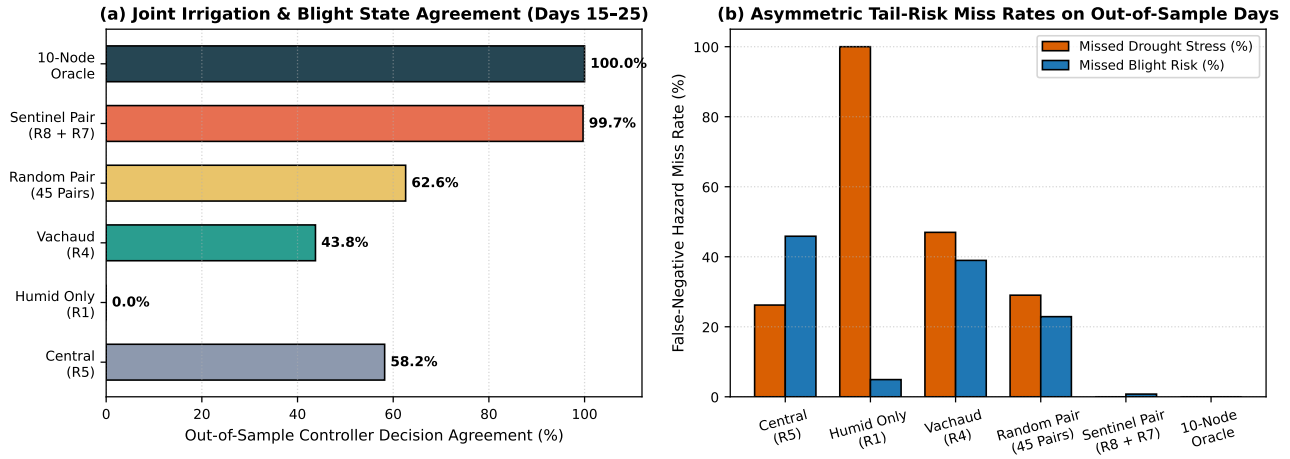


Figure 2: Out-of-Sample Phase 1 Controller Decision Agreement and Hazard Miss Rates (Days 15–25). (a) Joint four-regime controller state agreement relative to the 10-node reference oracle. (b) Asymmetric false-negative miss rates for dry-zone drought stress (VWC $\leq 22\%$) and humid-zone late-blight conducive conditions (RH $\geq 90\% \wedge$ LWS $\geq 50\%$).

Table 3: Phase 2 Randomized Complete Block Design (RCBD) Plot-Level Agronomic and Epidemiological Outcomes ($n = 4$ Blocks per Treatment; Mean $\pm$ SD Across Plots, Two-Way RCBD ANOVA with $\text{df}_{\text{trt}} = 3, \text{df}_{\text{err}} = 9$, and Pre-Specified 5% Yield Non-Inferiority Test)

Treatment Arm	Applied Water (mm)	Water Volume (m^3 ha^{-1})	Marketable Yield (t ha^{-1})	IWP (kg m^{-3})	Seasonal LWD (h)	AUDPC (%-days)	Final Blight Sev. (%)
T1: Conventional ET_c Reference	334.01 ± 2.34	3340.1 ± 23.4	33.02 ± 0.50	9.89 ± 0.14	1122.1 ± 56.0	451.8 ± 46.8	10.99 ± 1.20
T2: Single-Node Vachaud (R4)	255.34 ± 3.53	2553.4 ± 35.3	32.06 ± 0.21	12.56 ± 0.25	949.6 ± 51.8	273.1 ± 45.6	6.43 ± 1.16
T3: Two-Node Uncoupled (R8, R7)	250.08 ± 1.64	2500.8 ± 16.4	34.03 ± 0.59	13.61 ± 0.24	1025.0 ± 52.7	336.1 ± 41.2	8.15 ± 1.01
T4: Coupled Sentinel Controller	247.83 ± 5.32	2478.3 ± 53.2	34.40 ± 0.41	13.89 ± 0.47	977.1 ± 51.4	306.7 ± 44.0	7.27 ± 1.09
RCBD ANOVA $F_{3,9}$ (p-value)	569.69 ($< 10^{-9}$)	569.69 ($< 10^{-9}$)	110.11 (2.1×10^{-7})	437.42 ($< 10^{-9}$)	2624.9 ($< 10^{-12}$)	552.13 ($< 10^{-9}$)	461.52 ($< 10^{-9}$)
Standard Error of Diff. (SED)	2.464 mm	24.64 m^3 ha^{-1}	0.142 t ha^{-1}	0.123 kg m^{-3}	2.092 h	4.673 %-days	0.131%
$\text{LSD}_{0.05}$ (df = 9)	5.574 mm	55.74 m^3 ha^{-1}	0.322 t ha^{-1}	0.279 kg m^{-3}	4.732 h	10.571 %-days	0.295%

6 Limitations and Pre-Trial Field Execution Roadmap

While the digital-twin experiments validate the mathematical consistency, firmware state machine, and statistical power of the Sentinel Plant Network, physical field trials remain essential before claiming realized agronomic water savings or biological disease suppression:

1. **Site-Specific Dielectric Calibration:** Capacitive soil-moisture (‘GPIO5‘) and leaf-wetness (‘GPIO4‘) probes

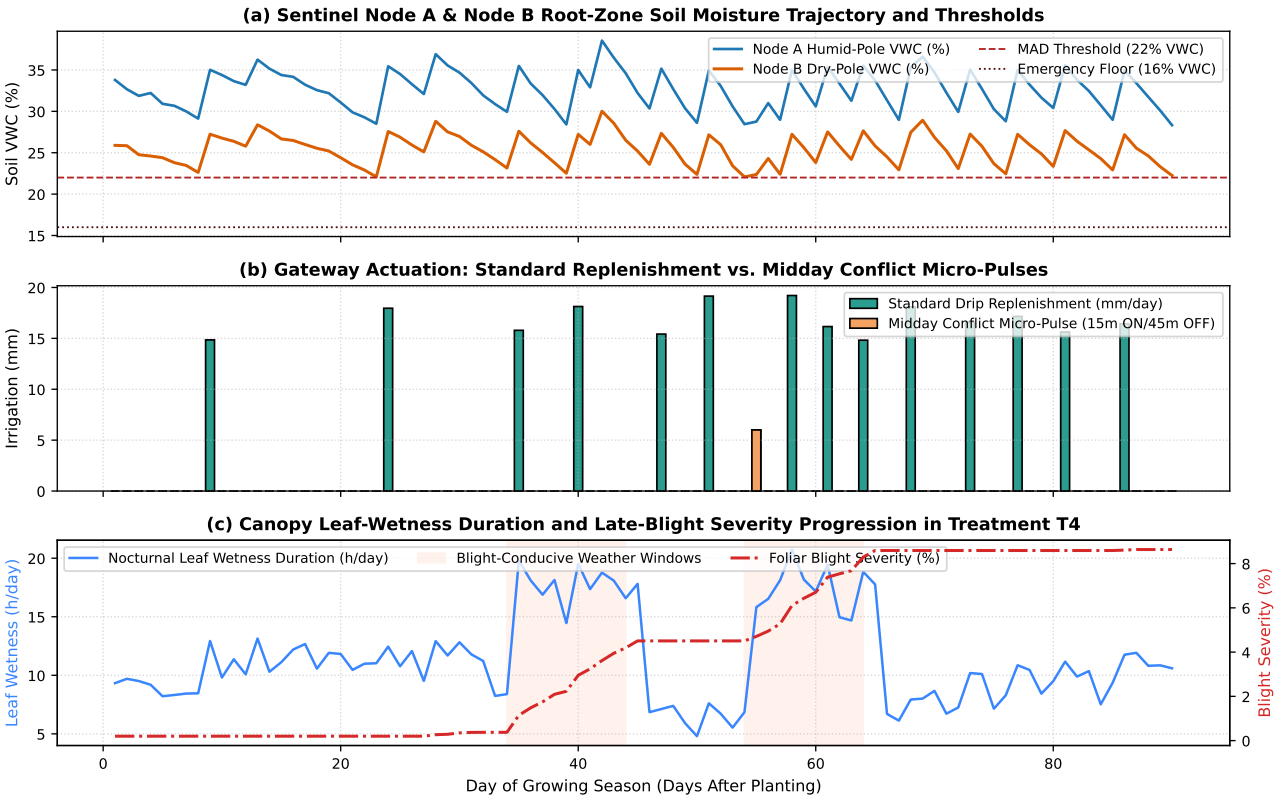


Figure 3: 90-Day Seasonal Operational Trace of the Coupled Sentinel Controller (T4, Block 1). (a) Dual-pole root-zone soil moisture trajectories at Node A (R8, humid pole) and Node B (R7, dry pole) relative to the 22% VWC management threshold and 16% VWC emergency floor. (b) Gateway valve actuation distinguishing standard drip replenishment from midday conflict micro-pulses (15 min ON / 45 min OFF inside 11:30–14:30) during days 34–44 and 54–64. (c) Realized nocturnal leaf-wetness duration and foliar late-blight progression.

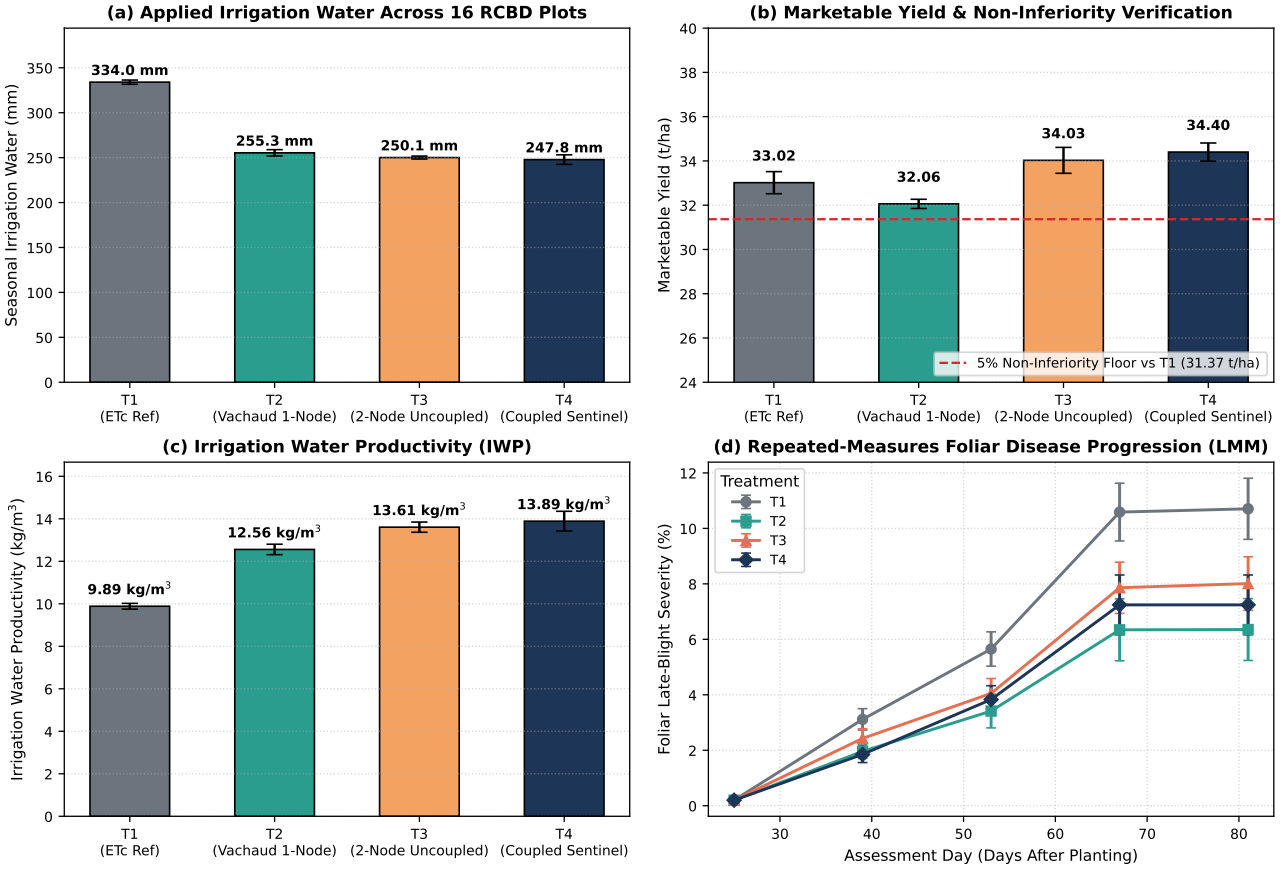


Figure 4: Phase 2 16-Plot RCBD Agronomic and Epidemiological Comparison Across Treatments T1–T4. (a) Seasonal applied irrigation water (mm). (b) Marketable yield (t ha^{-1}) relative to the pre-specified 5% non-inferiority floor (31.37 t ha^{-1} vs. T1). (c) Irrigation Water Productivity (IWP, kg m^{-3}). (d) Repeated-measures foliar blight severity trajectory across five bi-weekly scouting epochs.

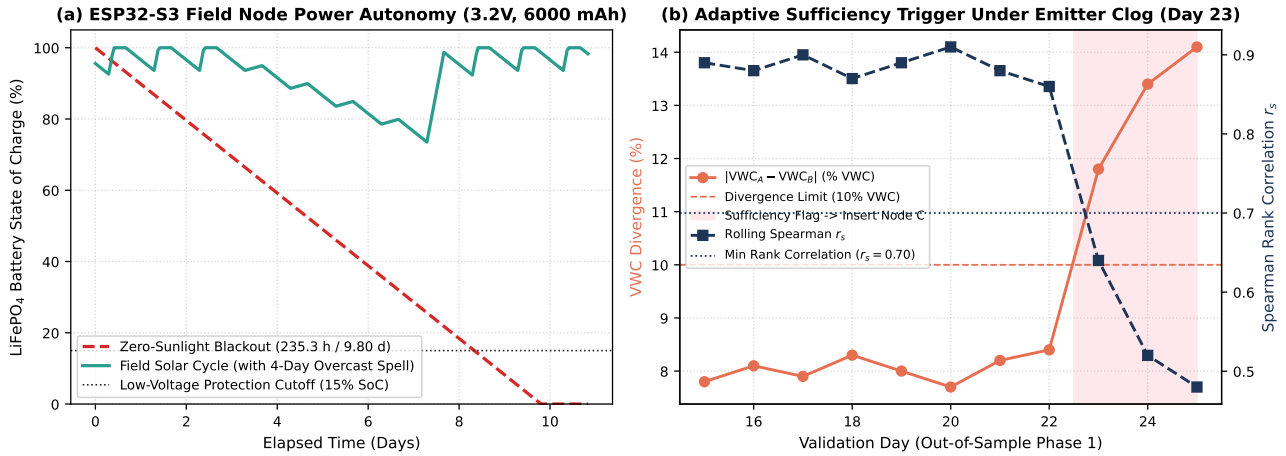


Figure 5: Hardware Power Autonomy and Adaptive Spatial Sufficiency Verification. (a) Simulated $LiFePO_4$ battery state-of-charge (SoC) trajectory under a complete zero-sunlight blackout (reaching 0% at 235.3 h = 9.80 days) versus normal diurnal 3 W solar recharging with a 4-day overcast spell (maintaining SoC > 62%). (b) Adaptive spatial sufficiency monitor detecting a simulated emitter clog on Day 23 when $|\mathrm{VWC}_A - \mathrm{VWC}_B|$ exceeds 10% VWC and rolling Spearman rank correlation drops below $r_s = 0.70$.

must undergo two-point gravimetric and filter-paper wetness calibration in the target field soil prior to Phase 1 deployment.

2. **Natural Inoculum Pressure Contingency:** As stipulated in Protocol Section 10, if natural *P. infestans* incidence in untreated check strips remains below 5% during the Phase 2 field trial, the trial will report sensor placement accuracy, water productivity, and canopy microclimate (LWD) outcomes without claiming biological disease suppression.

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